

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (EC) Examination

May 2017

EC 304/EC 304.01 Power Electronics

Date: 25.05.2017, Thursday

Time: 10:00 a.m. to 1:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a)** Define the following terms (i) Rectification efficiency (ii) Holding current (iii) Turn on and Turn off time. [03]
- (b)** Explain the architecture of offline UPS. [04]
- Q - 2 (a)** Derive the equation for Peak Reverse Recovery current flowing through the Power Diode. [05]
- (b)** Describe the principle of Phase controlled converter operation. [05]
- (c)** Explain the Control Characteristics of any two Power Devices. [04]

OR

- Q - 2 (a)** Explain the principle of three phase full converter having inductive load with necessary Diagrams. [05]
- (b)** Explain the function of freewheeling diode with the inductive load in controlled rectifiers. [05]
- (c)** Discuss the following with respect to BJT [04]
i) Secondary breakdown ii) Safe operating area
- Q - 3 (a)** Explain the methods by which a SCR can be turned on. [08]
- (b)** Explain the principle of single phase full converter having inductive load with necessary Diagrams. Also derive the expressions for average output voltage and RMS output voltage. [06]

OR

- (b)** Describe the switching characteristics of IGBT with neat figure of switching waveforms. [06]

SECTION – II

- Q - 4 (a)** Explain snubber circuits in brief [04]
- (b)** Which are the specifications of Power supplies? [03]
- Q - 5 (a)** The step down chopper has a resistive load of $R=10$ ohms and the input voltage is 220V. When the chopper switch remains on, its voltage drop is 2V and the chopping frequency is 1 kHz. If the duty cycle is 50%, determine (a) the average output voltage V_a (b) the RMS output voltage V_o (c) the chopper efficiency, (d) the effective input resistance R_i of the chopper. [05]
- (b)** Explain the principle of Step up operation of DC choppers [05]
- (c)** Explain with necessary diagram and equations the ON OFF controller for ac to ac conversion [04]

OR

- (c)** An On-off AC voltage controller has a resistive load of $R=10$ ohms and the RMS input voltage is $V_s=120$ V, 60 Hz. The Thyristor switch is ON for $n=25$ cycles and is OFF for $m=75$ cycles. Determine (a) the RMS output voltage V_o (b) the input power factor and (c) the average and the RMS current of the thyristors. [04]
- Q - 6 (a)** A single phase ac voltage controller has a resistive load of $R=10$ ohms and the input voltage is $V_s=120$ V, 60 Hz. The delay angle of the thyristor T1 is $(\pi/2)$. Determine (a) the RMS of the output voltage V_o (b) the input power factor PF and (c) the average input current. [06]
- (b)** Explain the operation of 3-phase bridge inverter operating in 180 degree mode with necessary waveforms [08]

OR

- (b)** Explain the operation of 3-phase bridge inverter operating in 120 degree mode with necessary waveforms [08]
